

Qatar-9b

R-2543

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_241205 · created 2026-08-26T00:15:03+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460649.8546 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.015
Transit depth (Rp/Rs)^2	2.92 +/- 0.5 [%]
Orbital Inclination (inc)	89.99 +/- 1.37
Airmass coefficient 1 (a1)	0.94 +/- 0.022
Airmass coefficient 2 (a2)	0.04 +/- 0.013
Scatter in the residuals of the lightcurve fit is	2.32 %
Best Comparison Star	#3 - [209, 278]
Optimal Method	PSF photometry
Transit Duration (day)	0.0788 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.171 ± 0.015 vs 0.1489 (deviation 1.23σ)
Duration fit vs published	0.0788 d vs 0.0779 d (deviation 0.46σ)
Mid-time O-C	-1.3 min
Depth SNR	9.5
Residual scatter	2.32 %

Light curve

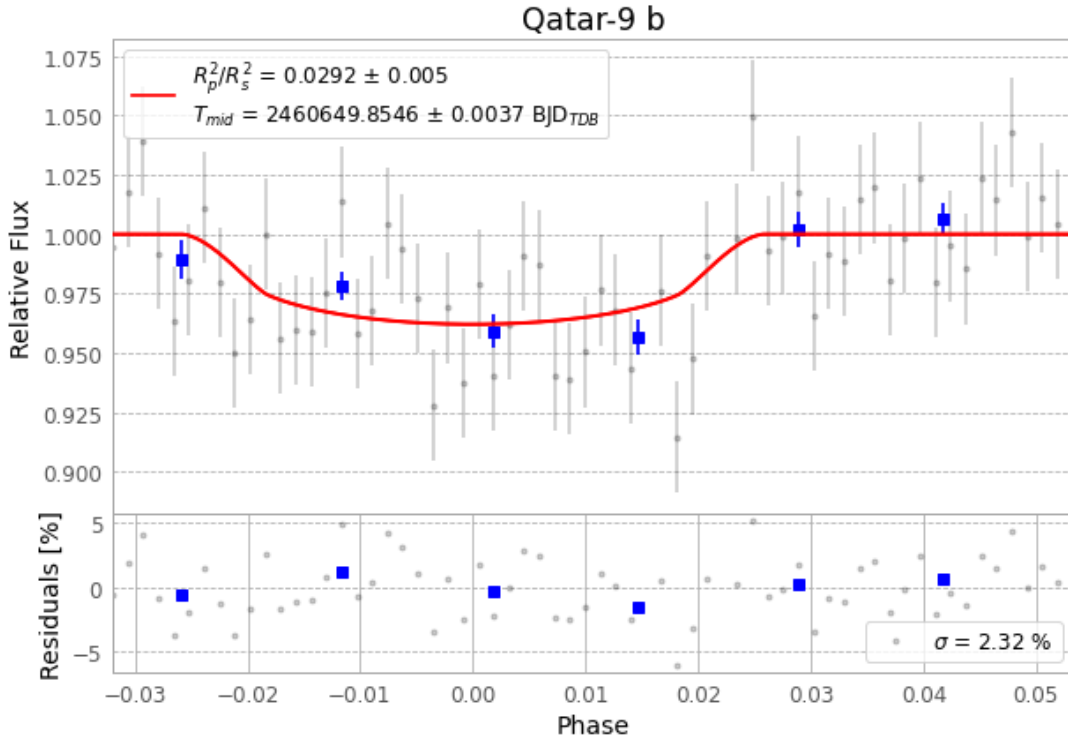


Figure 1 — FinalLightCurve_Qatar-9 b_2024-12-05.png (SHA-256 93dceb89e8e80632...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [416, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1b00fcd07da4278a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), Qatar-9241205071515.FITS → Qatar-9241205102415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-241205.log (0da09cf05532...), FinalLightCurve_Qatar-9 b_2024-12-05.png (93dceb89e8e8...), FinalLightCurve_Qatar-9 b_2024-12-05.csv (1da07ff73faf...)

Compute resources

Wall time 112.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:15Z.